Amphora Phase-0 Founder Briefing

System, Vision, Verifiable Results, and the IP Path to Own the Category

Internal — for the two founders. Blunt by design.

July 3, 2026

The one-paragraph truth

Phase 0 was not about winning a leaderboard. It was about building a prediction-and-formulation system whose every number survives a hostile audit, and proving it on public data with a protocol we hardened by red-teaming our own prior work. We found and fixed real landmines in that prior work — a benchmark whose metadata column literally *is* the label, a scaffold split that dumped 14% of the data into one test bucket, and a "TabPFN" gate model that was trained on *fake* soft labels. What we can now say honestly is worth more than what the last version claimed loudly. The moat is the discipline plus the proprietary data flywheel, not a single accuracy figure.

1 What we built (system design)

Two stages, one discipline.

1. **Multi-property prediction engine.** Feature-block ensembles (polymer-domain descriptors + cheminformatics descriptors + circular fingerprints + pretrained polymer embeddings) stacked out-of-fold into a meta-learner. Two builds are maintained on purpose: a COMMERCIAL-CLEAN stack (Apache/BSD/MIT only — the product) and a CEILING stack that adds non-commercial foundation models to measure headroom.

2. **Formulation compiler (the existing mvp19 engine).** Envelope → constraint set → feasibility-by-construction on the mix/solids surface → fail-closed gates → ranked, laboratory-executable blueprint with a qualification/acceptance matrix and a robust co-lead set. 23/23 regression tests green; naval lead unchanged.

Everything is reproducible from one command (`predict/verify.sh`) off frozen, hash-committed inputs.

2 Vision statement

A user states many specifications for a desired material; we screen the entire accessible chemical space faster than anyone, predict every relevant behavior with calibrated uncertainty, and hand back the best laboratory-ready candidates with an audit trail a customer's chemist and lawyer both accept. Coatings for the Navy prime is the wedge; the engine is property- and domain-general. Measured single-core screening is already 7,672 candidates/s; the fleet (384 cores, 3 GPUs) parallelizes linearly.

3 Verifiable Phase-0 results (the credible wins, and the honest non-wins)

T1 — polymer glass transition (7,367 polymers). Our commercial-clean stack vs. tuned gradient boosting, chemprop, and TabPFN-2/2.5 under identical frozen splits; paired cluster-bootstrap verdicts with pre-registered effect size. See Table 1/2. Headline is the Butina cluster split (hardest, most honest).

Table 1: T1 mean MAE (K).

Model	Random MAE (K)	Class-out MAE (K)	Butina MAE (K)
AFPE stack (ours, commercial-clean)	–	–	32.04
AFPE + TabPFN meta (ceiling; non-commercial)	21.10	30.36	–
Tuned GBM trio (LGBM+XGB+CatBoost)	23.25	33.48	–
TabPFN-2.5 zero-shot (non-commercial)	20.90	36.33	29.47
TabPFN-2 zero-shot (non-commercial)	22.64	30.55	30.88
Chemprop D-MPNN	25.28	42.44	36.15

Table 2: T1 paired verdicts vs. ours.

Family	Comparator	ΔMAE (K)	95% CI	Verdict
butina	TabPFN-2 zero-shot (non-commercial)	-1.16	[-2.47, +0.04]	TIE†

T2 — multi-property engine (5 properties). Evidence that one engine covers many behaviors; reported as a surrogate of molecular-dynamics simulation with replicate-noise floors (Table 3). We pre-declared that $n \approx 1$k favors TabPFN and report symmetrically.

Table 3: T2 MAE with replicate floors.
T2 results: NOT RUN

T3 — thermoset transfer (the property the customer needs). Honest negative: on 15 verified cured-network systems, transfer from polymer features does *not* beat the simple aromatic-fraction baseline (30.87°C grouped MAE). Cured-network T_g is under-determined by hardener alone. This is a wet-lab problem, not a modeling-cleverness problem — stated plainly so we do not sell a model we do not have.

T4 — rediscovery without cheating. Holding out five elite high-T_g hardeners and ranking them among property-matched decoys with a physics rigidity score: AUROC 0.7252, permutation p =0.008549572521373932. Real, significant enrichment — but it does *not* clear our pre-registered aggressive bar (BEDROC 0.168). We ship it as "significant signal, below discovery threshold," plus a 200-compound novel candidate slate (single CSV, every row MODELED).

4 Current IP inventory

- **System/method (mvp19):** envelope→constraint-set compilation with feasibility-by-construction on the mix/solids surface; fail-closed evidence-tiered gate discipline; qualification/ acceptance-matrix generation from envelope + public spec; robust co-lead recommendation. Positioned narrow and honest (a defensible *system*, not a novel optimizer).
- **Prediction engine (Phase 0):** the leakage-hardened protocol + the commercial-clean feature-block stacking recipe + the frozen-split reproducibility bundle.

- **Data assets:** the curated cured-network T_g anchor set and the coating-component library with Hansen parameters — small but genuinely proprietary; no public equivalent.
- **Superseded / disclosed:** mvp18's C4/E2 gate model was trained on a 48-row synthetic grid with *fabricated* "TabPFN-style" soft labels (its own manifest admits it). We disclose and retire it; the real TabPFN comparisons in this report replace it.

5 Licensing analysis (decisive for the product)

TabPFN-2.5/2.6/3 model weights and Google's TabFM weights are **non-commercial**. They are legitimate *comparator bars* and a *ceiling* reference, but they **cannot ship in the product**. Two consequences: (1) the product path must be the commercial-clean stack (RDKit + gradient boosting + our features + our data), which is exactly what we built and benchmarked; (2) a negotiated Prior Labs commercial license is a board-level roadmap option if the ceiling gap justifies it. TabPFN-3 weights are additionally token-gated and were not runnable here — disclosed, not papered over.

6 IP roadmap to dominate the space

1. **Phase 1 (now):** the mvp19 compiler is the "commit to chemistry" stage; wire Phase-0 predictors in only where honestly validated, else keep labeled priors. File the narrow system/method claims.
2. **Phase 2 (the wet-lab year — the real moat):** every measured panel (pot-life, VOC, adhesion, salt-fog, cathodic disbondment, cured T_g in context) re-enters through the structured capture template, promotes MODELED claims to REAL, calibrates the thermoset model on *in-context* data no competitor has, and closes an active-learning loop over the formulation space. Proprietary measured coating data is the asset competitors cannot copy; the flywheel schema already exists in prior code.
3. **Phase 3:** the engine generalizes to additional property heads and additional envelopes under the same frozen-protocol discipline; the compounding IP is (measured data × calibrated multi-property engine × auditable compiler), defended by trade secret on parameters/data and narrow patents on the system.

7 The blunt asks

(1) Fund the 30/60/90-day wet-lab panel — it is the only lever that turns MODELED into REAL and is the entire Phase-2 moat. (2) Decide the Prior Labs commercial-license question against the measured ceiling gap. (3) Hold the line on the discipline: the reason this briefing is credible is that it reports its own losses.